

Algebra 1
Unit 6
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

Amar is saving for a trip to Costa Rica with his friends. He has estimated that his travel expenses will cost between \$1,800 and \$2,400 total. He has \$300 in savings, and he earns \$16.50 per hour at his job. He wants to know how many hours he must work to afford the trip.

1. Write an inequality to represent the situation.

$$1800 \leq 300 + 16.50x \leq 2400$$

2. Solve the inequality.

$$\begin{array}{r} 1800 \leq 300 + 16.50x \\ -300 \quad -300 \end{array}$$

$$\frac{1500}{16.50} \leq \frac{16.50x}{16.50}$$

$$90.91 \leq x$$

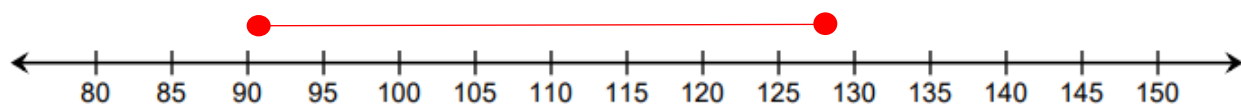
$$\begin{array}{r} 300 + 16.50x \leq 2400 \\ -300 \quad -300 \end{array}$$

$$\frac{16.50x}{16.50} \leq \frac{2100}{16.50}$$

$$x \leq 127.27$$

He must work between 90.91 and 127.27 hours to afford his trip.

3. Graph the solution.



4. Check any that apply.

- ☒ Solutions must be non-negative.
☐ Solutions must be integers.

5. Explain your selection(s) from question 4.

He can work fractions of hours, but cannot work negative hours.

Marcel just started her own business making cookies. She sells the cookies in boxes, charging \$15.00 per box. It costs her \$50.00 to buy the supplies. She wants to calculate how many boxes she must make to earn a profit of at least \$245.00.

6. Write an inequality to represent the situation.

$$15x - 50 \geq 245$$

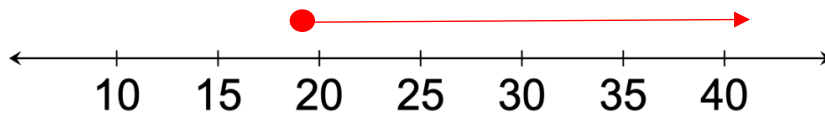
7. Solve the inequality.
- $$\begin{array}{r} 15x - 50 \geq 245 \\ +50 \quad +50 \end{array}$$

$$\frac{15x}{15} \geq \frac{295}{15}$$

$$x \geq 19.\bar{6}$$

She would need to sell at least 20 boxes to make a profit of at least \$245.00.

8. Graph the solution.



9. Check any that apply.

- ☒ Solutions must be non-negative.
- ☒ Solutions must be integers.

10. Explain your selection(s) from question 9.

She needs to sell whole boxes not fractions of boxes and she cannot sell negative boxes.